Assignment for Chapter 5 (B)

A. Fill-in-blank:

1. The reduction of the basic state table applies the concept of______________.

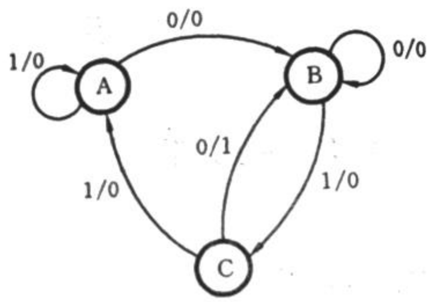
2. Let the number of states contained in the modified state table be n, and the number of flip-flops in the corresponding circuit is m, then the relationship between m and n should satisfy___________________。

3. Assume there is a Mealy type "0011" sequence detector, its modified state table consists of _________(how many) states, and its circuit contains ______(how many) flip-flops.

4. The state table of a synchronous sequential logic circuit is shown in the following table. If the initial state of the circuit is A and the input sequence of the circuit is x=010101, the corresponding output sequence of the circuit is_________________.

Present State	**Next state/Output**	
	x = 0	x= 1
A	B/0	C/1
B	C/1	B/0
C	A/0	A/1

5. The state diagram of a synchronous sequential logic circuit is shown in the following figure. If the initial state of the circuit is A, and under the input sequence 110101010, the corresponding output sequence of the circuit is _________________.



6. "In terms of the following equivalent pairs (A, B), (C, D) and (B, D), a maximum equivalent class {A, B, D} can be obtained." This statement is__(trule/ false)

7. To design a sequential logic circuit, the determination of the output function expression from the modified binary state table is independent of the type of the

selected flip-flop. This statement is ___________(trule/false).

B. Short Answer and Circuit Design：

Draw the Mealy and Moore type's basic state diagram of the "0101" sequence detector.

1. A typical input and output sequence is as follows:

Input x：1 1 0 1 0 1 0 1 0 0 1 1

Output Z：0 0 0 0 0 1 0 1 0 0 0 0

2. Reduce the following basic state table to a modified state table:

Present state	Next state/Output	
	X=0	X=1
A	B/0	C/0
B	A/0	F/0
C	F/0	G/0
D	A/0	C/0
E	A/0	A/1
F	C/0	E/0
G	A/0	B/1

3. Use the clock-controlled D, T, and J-K flip-flops as the storage devices of a synchronous sequential circuit, respectively, to implement the functions of the binary state table shown in the following table. Try to write the excitation function and the output function expressions; and try to answer “which type of flip-flop can achieve the simplest circuit?”